

equities_svi_analysis

August 4, 2026

```
[1]: # !pip install pytrends
      # !pip install nltk
```

```
[2]: # Import necessary libraries

import pandas as pd
import numpy as np
import re
import time
import matplotlib.pyplot as plt
import pytrends
import seaborn as sns

from pytrends.request import TrendReq
# import nltk

pd.set_option('future.no_silent_downcasting', True)
sns.set()
```

```
[3]: # nltk.download("punkt")
      # nltk.download("punkt_tab")
```

0.0.1 Fetching and Comparing Interest Over Time

```
[4]: pytrends = TrendReq(hl="en-US", tz=60) # Initialize TrendReq()

keywords = ["DANGOTE CEMENT", "GUARANTY TRUST BANK", "ZENITH BANK"]

timeframe = "today 5-y" # "today 12-m"

# pytrends.build_payload(kw_list=keywords, timeframe=timeframe)
pytrends.build_payload(kw_list=keywords, timeframe=timeframe, geo="NG") # Set
↳ to Nigeria Geographical Area

interest_over_time_df = pytrends.interest_over_time()
# interest_over_time_df = interest_over_time_df.fillna(False)
interest_over_time_df.query("isPartial == False", inplace=True)
```

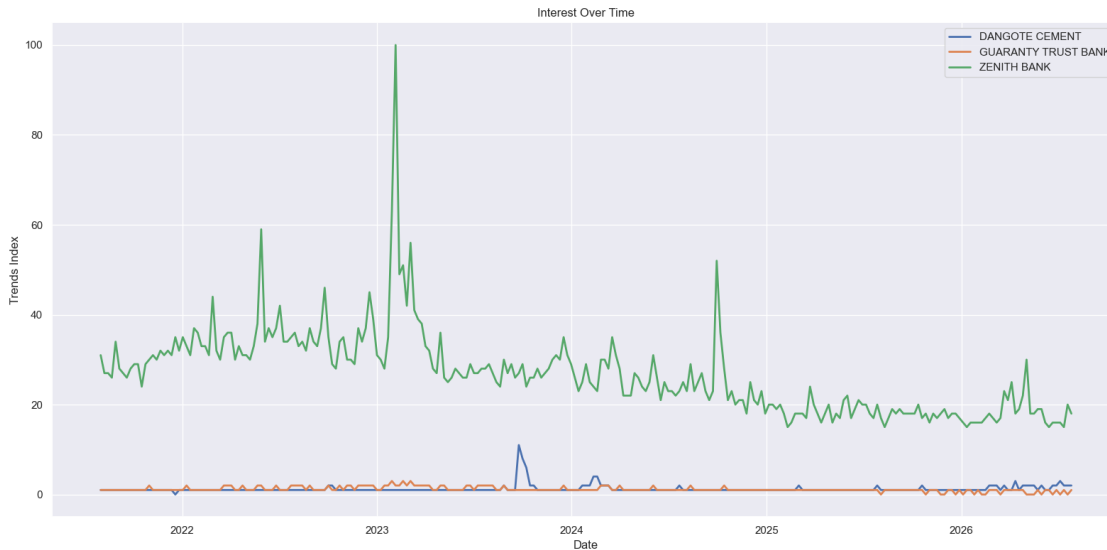
```
n = len(interest_over_time_df.columns) # Numeber of Columns
interest_over_time_df = interest_over_time_df.iloc[:, :n-1]
interest_over_time_df.to_csv("../outputs/SVI_Interest_Over_Time_NG.csv",
    ↪sep=",")
```

```
[5]: interest_over_time_df.head() # Overview of the dataset
```

```
[5]:
```

	DANGOTE CEMENT	GUARANTY TRUST BANK	ZENITH BANK
date			
2021-08-01	1	1	31
2021-08-08	1	1	27
2021-08-15	1	1	27
2021-08-22	1	1	26
2021-08-29	1	1	34

```
[6]: # Plotting the interest over time
plt.figure(figsize=(16, 8))
for keyword in keywords:
    sns.lineplot(
        x=interest_over_time_df.index,
        y=interest_over_time_df[keyword],
        label = keyword,
        linewidth=2
    )
plt.xlabel("Date")
plt.ylabel("Trends Index")
plt.title("Interest Over Time")
plt.legend(loc="upper right")
plt.grid(True)
plt.tight_layout()
plt.savefig("../outputs/SVI_Interest_Over_Time_NG.png", dpi=300,
    ↪bbox_inches="tight")
plt.show()
```



```
[7]: avg_SVI = interest_over_time_df.mean() # Average SVI
std_SVI = interest_over_time_df.std() # Std Dev SVI
max_SVI = interest_over_time_df.max() # Maximum SVI
min_SVI = interest_over_time_df.min() # Minimum SVI

summary_stats = pd.DataFrame({
    "Keyword": interest_over_time_df.columns,
    "Mean SVI": avg_SVI,
    "Std Dev SVI": std_SVI,
    "Max SVI (Peak)": max_SVI,
    "Min SVI": min_SVI
})
# summary["Average SVI"] = avg_SVI
# summary["Standard Deviation SVI"] = std_SVI
summary_stats.reset_index(drop=True, inplace=True)
summary_stats.to_csv("../outputs/SVI_Summary_Statistics.csv")
print("Summary Statistics for Search Volume Index:")
display(summary_stats)
```

Summary Statistics for Search Volume Index:

	Keyword	Mean SVI	Std Dev SVI	Max SVI (Peak)	Min SVI
0	DANGOTE CEMENT	1.229885	0.911902	11	0
1	GUARANTY TRUST BANK	1.153257	0.533024	3	0
2	ZENITH BANK	26.808429	9.404259	100	15

Correlation of Each Keywords

```
[8]: ##K Keywords Correlations
corr_interest_over_time = interest_over_time_df.corr()
```

```
corr_interest_over_time.to_csv("../outputs/SVI_Corr_Interest_Over_Time_NG.csv",
    ↪sep=",")
corr_interest_over_time
```

```
[8]:
```

	DANGOTE CEMENT	GUARANTY TRUST BANK	ZENITH BANK
DANGOTE CEMENT	1.000000	-0.096500	-0.082749
GUARANTY TRUST BANK	-0.096500	1.000000	0.576738
ZENITH BANK	-0.082749	0.576738	1.000000

Interest Over Time:

The Google Trends data shows that **Zenith Bank consistently dominates search interest in Nigeria** compared to Dangote Cement and Guaranty Trust Bank. From 2021 to 2026, Zenith Bank's trend line remains significantly higher, with noticeable peaks that suggest periods of heightened public attention—likely tied to product launches, service issues, or major announcements. In contrast, Dangote Cement and Guaranty Trust Bank maintain relatively low and stable search volumes, indicating less frequent spikes in public interest.

This suggests that **consumer-facing financial institutions attract more consistent online attention** than industrial companies like Dangote Cement, which may only trend during specific events (e.g., corporate news or policy changes).

0.0.2 Analyzing Interest by Region

```
[9]: pytrends.build_payload(kw_list=keywords, timeframe=timeframe, geo="NG") # Set
    ↪to Nigeria Geographical Area
interest_by_region_df = pytrends.interest_by_region()
interest_by_region_df.index.name = "REGION"
interest_by_region_df.to_csv("../outputs/SVI_Interest_By_Region_NG.csv",
    ↪sep=",")
```

Sort by DANGOTE CEMENT

```
[10]: interest_by_region_df.sort_values(by="DANGOTE CEMENT", ascending=False).head()
```

```
[10]:
```

	DANGOTE CEMENT	GUARANTY TRUST BANK	ZENITH BANK
REGION			
Kogi	13	2	85
Benue	10	1	89
Gombe	6	3	91
Ekiti	6	4	90
Osun	6	2	92

Note:

Dangote Cement registers modest interest, with its highest values in Kogi and Benue.

Sort by GUARANTY TRUST BANK

```
[11]: interest_by_region_df.sort_values(by="GUARANTY TRUST BANK", ascending=False).
      ↪head()
```

```
[11]:
```

	DANGOTE CEMENT	GUARANTY TRUST BANK	ZENITH BANK
REGION			
Adamawa	3	4	93
Lagos	4	4	92
Ogun State	5	4	91
Federal Capital Territory	3	4	93
Ekiti	6	4	90

```
[12]: min_, max_ = interest_by_region_df["GUARANTY TRUST BANK"].min(),
      ↪interest_by_region_df["GUARANTY TRUST BANK"].max()
print("Minimum search score:", min_)
print("Maximum search score:", max_)
```

Minimum search score: 0

Maximum search score: 4

Note:

Guaranty Trust Bank appears weaker in regional searches, often scoring below 5 in most states.

Sort by ZENITH BANK

```
[13]: interest_by_region_df.sort_values(by="ZENITH BANK", ascending=False).head()
```

```
[13]:
```

	DANGOTE CEMENT	GUARANTY TRUST BANK	ZENITH BANK
REGION			
Yobe	1	0	99
Taraba	1	1	98
Zamfara	2	0	98
Akwa Ibom	3	1	96
Abia	3	1	96

Note:

Zenith Bank shows overwhelming dominance in states such as Yobe, Taraba, and Anambra, with values above 90 in relative interest.

Interest by Region:

Regional analysis highlights strong variation across Nigerian states:

- **Zenith Bank** shows overwhelming dominance in states such as Yobe, Taraba, and Anambra, with values above 90 in relative interest.
- **Dangote Cement** registers modest interest, with its highest values in Kogi and Benue.
- **Guaranty Trust Bank** appears weaker in regional searches, often scoring below 5 in most states.

0.0.3 Analyzing Related Queries (Top and Rising)

```
[16]: pytrends.build_payload(kw_list=keywords, timeframe=timeframe, geo="NG") # Set_
      ↪to Nigeria Geographical Area
      related_query_dict = pytrends.related_queries()
```

For Dangote Cement

```
[17]: display(related_query_dict["DANGOTE CEMENT"]["top"])
```

None

```
[18]: display(related_query_dict["DANGOTE CEMENT"]["rising"])
```

None

For Guaranty Trust Bank

```
[19]: display(related_query_dict["GUARANTY TRUST BANK"]["top"])
```

None

```
[20]: display(related_query_dict["GUARANTY TRUST BANK"]["rising"])
```

None

For Zenith Bank

```
[21]: top = related_query_dict["ZENITH BANK"]["top"]
      top.rename(columns={"query": "Query", "value": "Value"}, inplace=True)
      top.to_csv("../outputs/SVI_Zenith_Top_Related_Queries.csv", sep=",")
      display(top.head())
```

	Query	Value
0	zenith bank code	100
1	zenith internet banking	30
2	zenith bank ussd	30
3	zenith bank internet banking	30
4	zenith bank internet	30

Note: > Top queries include “zenith bank code”, “zenith bank customer care”, and “zenith bank internet”.

```
[22]: rising = related_query_dict["ZENITH BANK"]["rising"]
      rising.rename(columns={"query": "Query", "value": "Value"}, inplace=True)
      top.to_csv("../outputs/SVI_Zenith_Rising_Related_Queries.csv", sep=",")
      display(rising.head())
```

	Query	Value
0	what is hardware token in zenith bank	550
1	ussd for zenith bank	180
2	ussd code for zenith bank	170
3	zenith bank shares	140
4	zenith ussd code	140

Note: > Rising queries emphasize digital access issues such as “zenith bank customer care number 24 hours” and “zenith bank network today”. —

Summary of Related Queries

For **Zenith Bank**, related queries are heavily directed toward **customer service and digital banking tools**:

- Top queries include “zenith bank code”, “zenith bank customer care”, and “zenith bank internet”.
- Rising queries emphasize digital access issues such as “zenith bank customer care number 24 hours” and “zenith bank network today”.

However, Dangote Cement and Guaranty Trust Bank return no top and rising queries

0.1 Overall Interpretation

Overall, the analysis shows:

- **Zenith Bank** dominates both national and regional search interest, driven by customer service and digital banking queries.
 - **Dangote Cement** and **Guaranty Trust Bank** attract far less attention, with interest concentrated in specific regions.
 - The data underscores the **importance of digital accessibility and customer support** in shaping public search behavior for Nigerian banks.
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